

ORIGINAL RESEARCH

Renin, Aldosterone, and Treatment Response to Dapagliflozin, Spironolactone, and Their Combination

An Analysis From SOGALDI-PEF

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ABSTRACT

BACKGROUND Hypertension and heart failure with preserved ejection fraction (HFpEF) often coexist, and sodium retention is a common feature of both conditions. Renin levels are suppressed in the setting of sodium overload and may modulate the blood pressure (BP) response to mineralocorticoid receptor antagonists (MRAs). The role of renin and aldosterone, and their influence on the response to dapagliflozin, spironolactone, and their combination, has not been tested in HFpEF.

OBJECTIVES The aim of this study was to assess the role of renin and aldosterone in patients with HFpEF enrolled in the SOGALDI-PEF (Dapagliflozin With or Without Spironolactone for HFpEF) trial.

METHODS Plasma renin and aldosterone levels were measured at baseline and at each study visit within a crossover design. Two main sequences were evaluated: dapagliflozin alone and spironolactone/dapagliflozin combination, each administered for 12 weeks.

RESULTS A total of 106 patients were included. The median (percentile 25-75) renin level at baseline was 29.7 mU/L (17.6-61.6 mU/L); tertile 1 comprised patients with renin levels ≤ 21 mU/L. Compared with patients with higher renin (tertile 2/tertile 3), those with lower levels had higher BP (systolic blood pressure/diastolic blood pressure), serum sodium, lateral E/e', pulse wave velocity, and peripheral edema. Renin levels, but not aldosterone levels, modified the effect on BP: compared with dapagliflozin alone, the spironolactone/dapagliflozin combination reduced systolic blood pressure by -10.9 mm Hg (95% CI: -16.4 to -5.4 mm Hg) among patients with low renin (tertile 1) but had no significant effect at higher renin levels: tertile 2, -5.6 mm Hg (95% CI: -11.3 to 0.02 mm Hg); and tertile 3, 1.5 mm Hg (95% CI: -4.1 to 7.0 mm Hg) (interaction P trend = 0.007). Echocardiographic parameters and pulse wave velocity followed a similar pattern. The correlation between plasma renin and aldosterone was poor ($\rho = 0.15$; $P = 0.12$).

CONCLUSIONS Low plasma renin levels (≤ 21 mU/L) identify a high BP, sodium overload state and predict a more pronounced BP reduction with the spironolactone/dapagliflozin combination compared with dapagliflozin alone. Additional findings support a favorable impact of spironolactone/dapagliflozin in cardiovascular structure and function in patients with low renin levels. Aldosterone levels did not predict treatment response. (Dapagliflozin With or Without Spironolactone for HFpEF [SOGALDI-PEF]; [NCT05676684](https://doi.org/10.1016/j.jchf.2026.103111)) (JACC Heart Fail. 2026;■:103111) © 2026 by the American College of Cardiology Foundation.

ABBREVIATIONS
AND ACRONYMS**ARR** = aldosterone-to-renin ratio**BNP** = brain natriuretic peptide**BP** = blood pressure**HF** = heart failure**HFpEF** = heart failure with preserved ejection fraction**DBP** = diastolic blood pressure**eGFR** = estimated glomerular filtration rate**MRA** = mineralocorticoid receptor antagonist**NT-proBNP** = N-terminal pro-B-type natriuretic peptide**PWV** = pulse wave velocity**SBP** = systolic blood pressure

The SOGALDI-PEF (Dapagliflozin With or Without Spironolactone for HFpEF; [NCT05676684](#)) trial showed that the spironolactone/dapagliflozin combination was superior to dapagliflozin alone for reducing N-terminal pro-B-type natriuretic peptide (NT-proBNP) levels and blood pressure (BP) among patients with heart failure with preserved ejection fraction (HFpEF). However, this combination was also associated with a more pronounced reduction in estimated glomerular filtration rate (eGFR) and increases in serum potassium compared with dapagliflozin alone.¹ Because mineralocorticoid receptor antagonists (MRAs) can cause adverse effects such as hyperkalemia and worsening kidney function, there is a need to identify patients who are most likely to derive benefit from MRA therapy while minimizing adverse effects.²

Plasma renin levels are a surrogate marker for sodium retention, as patients with higher sodium intake or sodium avidity typically exhibit lower renin concentrations.³ In addition to sodium excess, renin is suppressed in states of autonomous aldosterone secretion, such as primary hyperaldosteronism.⁴ Plasma renin has been shown to predict the antihypertensive response to MRAs, with the most pronounced BP reductions observed in patients with low renin levels.^{5,6} Moreover, aldosterone breakthrough is common in populations with cardiovascular-kidney-metabolic diseases, but there has been little evaluation of these biomarkers in HFpEF populations.⁷

Patients with HFpEF commonly exhibit neurohormonal activation, leading to sodium retention and extracellular volume expansion that manifests as congestive signs and symptoms.⁸ However, natriuresis and diuresis in response to volume expansion are often blunted because of an impaired renal

response to endogenous natriuretic peptides, perpetuating the cycle of sodium avidity and volume overload.⁹ In HFpEF, MRAs reduce HF hospitalizations, decrease diuretic requirements, improve signs and symptoms of congestion, and lower BP, supporting their role in targeting HF-related volume overload and its deleterious consequences.¹⁰⁻¹⁴ However, whether baseline plasma renin and aldosterone levels influence the response to MRAs in HFpEF remains unknown, as these hormones have not been measured in HFpEF randomized outcome trials.

Hypertension and HFpEF often coexist, and sodium retention is a common feature in both conditions; therefore, a more pronounced BP response to therapies targeting sodium retention would also be expected in HFpEF at low renin levels. To better clarify the role of renin and aldosterone in HFpEF, we have measured these hormones at baseline and at subsequent study visits in the SOGALDI-PEF trial. Our goals were to: 1) study the associations of renin and aldosterone with patient characteristics; 2) assess how spironolactone, dapagliflozin, and their combination affect renin and aldosterone levels; and 3) determine whether renin and aldosterone levels modify the response to dapagliflozin, spironolactone, and the dapagliflozin/spironolactone combination.

METHODS

The design, methods, and main results of SOGALDI-PEF have been published previously.^{1,15} Briefly, SOGALDI-PEF is an investigator-initiated and investigator-led 2-center prospective randomized open-label, blinded endpoint crossover trial aimed at comparing 3 sequences (dapagliflozin, spironolactone, or the dapagliflozin/spironolactone combination) for 3 periods of 12 weeks in patients with HF with mid-range EF and with HFpEF. The trial is registered on ClinicalTrials.gov ([NCT05676684](#))

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*A list of the SOGALDI-PEF investigators is provided in the [Supplemental Appendix](#).

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The authors attest they are in compliance with human studies committees and animal welfare regulations of the authors' institutions and Food and Drug Administration guidelines, including patient consent where appropriate. For more information, visit the [Author Center](#).

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and EudraCT (No. 2020-000973-26) and was approved by the CEIC (Portuguese Ethics Committee for Clinical Research) and the INFARMED (National Authority of Medicines and Health Products). The study was conducted in accordance with Good Clinical Practice and the Declaration of Helsinki.

Patient enrollment started in September 2022; in July 2023, we submitted a protocol amendment to suppress the spironolactone alone arm and to shorten the washout period from 4 weeks to 1 week, after 2 independent trials confirmed the benefit of sodium-glucose cotransporter 2 inhibitors in this population.¹⁵⁻¹⁷ Recruitment was completed in June 2024, and the last patient last visit was held in November 2024.

The primary outcome was the difference in circulating levels of log-transformed NT-proBNP between the dapagliflozin/spironolactone combination vs dapagliflozin alone. Treatment sequences were blinded to the investigators analyzing the primary endpoint. Circulating NT-proBNP levels were centrally measured in triplicate using an immunoassay (Alinity i NT-proBNP, #4S7920, Abbott Diagnostics) performed on an automated laboratory system (Alinity ci-series, Abbott Core Laboratory).

ENTRY CRITERIA. SOGALDI-PEF enrolled adult (aged ≥ 50 years) outpatients with symptomatic HF (NYHA functional class II to IV), a left ventricular ejection fraction $>40\%$, and evidence of structural heart disease (left atrial enlargement, left ventricular hypertrophy, or features of diastolic dysfunction). All patients were required to have elevated natriuretic peptide levels at screening (≤ 12 weeks before enrollment), with eligibility thresholds adjusted based on the presence of atrial fibrillation on electrocardiogram. For patients in sinus rhythm, levels of NT-proBNP ≥ 220 pg/mL or brain natriuretic peptide (BNP) ≥ 80 pg/mL were required. For patients in atrial fibrillation, the NT-proBNP and BNP thresholds were tripled (ie, ≥ 660 pg/mL and ≥ 240 pg/mL, respectively). An eGFR ≥ 30 mL/min/1.73 m² and a serum potassium level ≤ 5.5 mmol/L at screening were also required for inclusion. Patients could not have been treated with sodium-glucose cotransporter 2 inhibitors or MRAs nor have been hospitalized for HF decompensation within the previous 30 days before inclusion.

DAPAGLIFLOZIN AND SPIRONOLACTONE DOSAGE. The dapagliflozin dose was 10 mg/d without dose adjustments. The spironolactone dose varied from 25 mg every other day to 25 mg/d based on eGFR and serum potassium levels.¹⁸ Spironolactone was initiated at 25 mg every other day if eGFR

was <60 mL/min/1.73 m² or if eGFR was ≥ 60 mL/min/1.73 m² with serum potassium levels between 5.0 and 5.5 mmol/L. Spironolactone was initiated at a dose of 25 mg/d if eGFR was ≥ 60 mL/min/1.73 m² and serum potassium levels were <5.0 mmol/L. Additional spironolactone dose adjustments could be performed throughout the trial at the investigator's discretion. Forty-seven (45%) patients who underwent spironolactone sequences received 25 mg every other day. The weighted median spironolactone dose during the trial was approximately 19 mg/d.

STUDY VISITS AND PROCEDURES. Follow-up included scheduled visits at baseline/randomization (visit 1) and at the end of each sequence. Planned standard of care appointments were maintained and conducted according to the physician's discretion. A telephone follow-up was performed at the beginning of each new sequence to ensure that the patient understood the protocol and the upcoming study drug sequence and to monitor adverse events. Blood and spot urine measurements were performed serially at study visits. Temporary interruption of study drug and reassessment of potassium levels within 7 to 15 days were recommended if serum potassium levels were ≥ 5.5 mmol/L, but the decision was left to the discretion of the investigators.

QUANTIFICATION OF PLASMA RENIN AND ALDOSTERONE. Venous blood samples were collected into EDTA tubes in the morning after 5 minutes with the patient in the supine position; samples were immediately centrifuged (3,000 rpm, 10 minutes, room temperature), and plasma was aliquoted and stored at -80°C until analysis. Baseline samples were collected before initiation of the first treatment sequence ($n = 106$). End-of-treatment samples were obtained on the last day of each period for dapagliflozin alone ($n = 103$), the dapagliflozin/spironolactone combination ($n = 103$), and spironolactone monotherapy ($n = 39$).

Plasma renin concentration was measured by using a direct chemiluminescent immunoassay (MAGLUMI Direct Renin, #130206511M, Snibe Diagnostics) on a fully automated MAGLUMI analyzer, according to the manufacturer's instructions. Plasma aldosterone was quantified by using a competitive chemiluminescent immunoassay (MAGLUMI Aldosterone, #130256007M, Snibe Diagnostics). Renin and aldosterone were measured in the same plasma sample, and all samples were analyzed in a single batch to minimize interassay variability.

STUDY OUTCOMES. The primary outcome in the current study was the difference in systolic blood pressure (SBP) between the spironolactone/

dapagliflozin combination and dapagliflozin sequences across renin levels (ie, treatment-by-renin interaction for SBP difference). This outcome was chosen based on prior data showing that renin levels modified the effect of spironolactone on BP, whereby the effect to lower SBP was most pronounced at low renin levels.^{5,6}

BP was measured in the sitting position with 3 measures for each patient/visit in a quiet room in the presence of a physician or a nurse, and the average of the 3 measurements was taken. Secondary outcomes included between-treatment sequence differences in diastolic blood pressure (DBP), NT-proBNP, left atrial diameter, left atrial volume, left ventricular mass, lateral E/e', E/A ratio, left ventricular ejection fraction, pulmonary artery systolic pressure, pulse wave velocity (PWV), eGFR, urinary albumin-to-creatinine ratio, serum potassium, body weight, spot urinary sodium-to-creatinine ratio, spot urine sodium, and spot urine potassium, with respective treatment-by-renin, aldosterone, and aldosterone-to-renin ratio (ARR) interactions.

STATISTICAL ANALYSIS. Baseline levels of plasma renin, aldosterone, and ARR were divided in “data-driven” tertiles (T1, T2, and T3) with a similar distribution of patients within each tertile. Patient characteristics were described across tertiles of renin, aldosterone, and ARR using descriptive statistics and respective statistical trend tests. The effect of treatment sequences on renin, aldosterone, and ARR levels was assessed by using a multilevel mixed model for repeated measures, including treatment sequence, sequence order, treatment period, and center as fixed effects, along with the global baseline value of the outcome (ie, not period-specific). The model included random intercepts for subjects (patient identifier), with observations nested within subjects, and an unstructured covariance matrix to account for within-subject correlations. Models were fitted by using restricted maximum likelihood.

A ratio of geometric means using a robust variance estimator was used to report relative ratios of log-transformed variables. To test whether renin, aldosterone, and ARR levels modified the effect of SBP and other studied outcomes, a similar mixed model for repeated measures model was fit but with the studied outcomes as dependent variables, and treatment sequence, baseline renin, aldosterone, and ARR levels, and respective treatment-by-hormone interaction in the model. The interaction test was an ordered interaction P trend test to account for the ordered categories of the variables (tertiles). To explore which variables were associated with the

odds of having low renin levels at baseline, a logistic regression model was fit with low renin (T1) as outcome variable, and all variables from Table 1 with a value of $P < 0.15$ were inserted in the full model as independent variables; a stepwise value of $P < 0.05$ was then used for entering the model. Subgroup analyses were performed among patients with and without BP $>130/80$ mm Hg at baseline and with and without ≥ 3 antihypertensive medications at baseline. As previously reported, treatment-by-period interaction tests were statistically nonsignificant, supporting the absence of significant carryover effects.¹

Analyses were performed by using Stata version 18 (StataCorp LLC). A value of $P < 0.05$ was considered statistically significant. Missing data were not imputed. Analyses were conducted using all available observations, providing a valid inference under a missing-at-random assumption. The proportion of missing data for each outcome across treatment periods and sequences is reported in the tables. The lower number of observations in the spironolactone monotherapy group reflects the mid-trial protocol amendment and not a systematic exclusion. No multiple test correction was performed given the exploratory nature of the analyses.

RESULTS

PATIENT CHARACTERISTICS ACROSS RENIN AND ALDOSTERONE LEVELS. A total of 106 patients had plasma renin and aldosterone levels determined at baseline. The median plasma renin level at baseline was 29.7 mU/L (25th-75th percentile: 17.6-61.6 mU/L). The corresponding tertiles of plasma renin were: tertile 1 (T1) ($n = 36$), 12.4 mU/L (25th-75th percentile: 6.7-18.0 mU/L); tertile 2 (T2) ($n = 35$), 30.0 mU/L (25th-75th percentile: 24.0-36.4 mU/L); and tertile 3 (T3) ($n = 35$), 96.1 mU/L (25th-75th percentile: 61.6-211.0 mU/L). Compared with patients with higher renin levels (T2 and T3), those with lower levels (T1 [ie, renin levels ≤ 21 mU/L]) had higher SBP, DBP, serum sodium, spot urine sodium excretion, lateral E/e', and PWV. They also presented more frequent peripheral edema, despite no differences in eGFR, which tended to be higher at lower renin levels. Furthermore, no statistically significant differences across renin tertiles were observed regarding age, sex, diabetes status, NT-proBNP, plasma aldosterone levels, or background medication such as diuretics or angiotensin-converting enzyme inhibitors/angiotensin receptor blockers (Table 1).

The median plasma aldosterone level at baseline was 68.1 pg/mL (25th-75th percentile: 41.7-115 pg/mL). The corresponding tertiles of plasma aldosterone

TABLE 1 Baseline Patient Characteristics According to Tertiles of Renin Levels

	Renin T1	Renin T2	Renin T3	P Value
Plasma renin, mU/L	12.4 (6.7-18.0); n = 36	30.0 (24.0-36.4); n = 35	96.1 (61.6-211.0); n = 35	—
Age, y	77.0 (73.5-82.5)	74.0 (68.0-80.0)	75.0 (71.0-80.0)	0.12
Female	22 (61)	18 (51)	21 (60)	0.67
HF etiology				0.91
Hypertensive	21 (58)	22 (63)	19 (54)	
Ischemic	5 (14)	6 (17)	6 (17)	
Other/unknown	10 (28)	7 (20)	10 (29)	
Prior HFH	6 (17)	5 (14)	8 (23)	0.63
MI	3 (8)	4 (11)	4 (11)	0.88
Stroke	8 (22)	7 (20)	6 (17)	0.87
CKD	10 (28)	18 (51)	16 (46)	0.11
AF ECG	17 (47)	12 (34)	14 (40)	0.54
Hypertension	33 (92)	34 (97)	30 (86)	0.23
BP >130/80 mm Hg	27 (75)	16 (46)	13 (37)	0.004
Diabetes	15 (42)	15 (43)	19 (54)	0.50
HbA _{1c} , %	6.0 (5.6-6.4)	5.9 (5.6-6.4)	6.0 (5.7-6.5)	0.73
BMI, kg/m ²	28.9 ± 5.2	30.0 ± 6.3	29.6 ± 6.0	0.72
Heart rate, beats/min	66.4 ± 10.9	70.3 ± 15.5	66.6 ± 13.1	0.40
SBP, mm Hg	139.7 ± 17.2	132.2 ± 18.8	128.1 ± 19.6	0.030
DBP, mm Hg	76.3 ± 8.8	73.8 ± 11.6	69.3 ± 10.2	0.017
Pulmonary rales	20 (56)	23 (66)	20 (57)	0.65
Peripheral edema	29 (81)	28 (80)	20 (57)	0.042
JVD	24 (67)	25 (71)	25 (71)	0.88
Orthopnea	4 (11)	5 (14)	4 (11)	0.90
NYHA functional class				0.99
II/III	31 (86)	30 (86)	30 (86)	
IV	5 (14)	5 (14)	5 (14)	
Angina pectoris	4 (11)	2 (6)	0 (0)	0.13
ACEI/ARB	26 (72)	21 (60)	25 (71)	0.47
Beta-blocker	26 (72)	24 (69)	24 (69)	0.93
Thiazide	2 (6)	8 (23)	3 (9)	0.061
CCB	19 (53)	9 (26)	16 (46)	0.057
Furosemide	25 (69)	22 (63)	26 (74)	0.58
≥3 antihypertensive drugs	21 (58)	16 (46)	20 (57)	0.50
Statin	29 (81)	32 (91)	32 (91)	0.27
GLP1-RA	2 (6)	5 (14)	7 (20)	0.19
Hemoglobin, g/dL	12.6 ± 2.0	13.1 ± 1.7	12.7 ± 1.7	0.55
Urea, mg/dL	45.0 ± 16.7	52.3 ± 27.6	63.5 ± 39.5	0.032
Creatinine, mg/dL	1.0 ± 0.3	1.0 ± 0.3	1.2 ± 0.5	0.12
eGFR, mL/min/1.73 m ²	74.1 ± 20.3	70.2 ± 22.4	63.9 ± 22.6	0.14
eGFR <60 mL/min/1.73 m ²	9 (25)	11 (31)	15 (43)	0.27
Sodium, mmol/L	140.8 ± 2.2	139.4 ± 2.5	138.7 ± 3.8	0.008
Potassium, mmol/L	4.3 ± 0.4	4.2 ± 0.5	4.4 ± 0.5	0.16
Uric acid, mg/dL	5.4 ± 1.7	6.0 ± 2.1	6.9 ± 2.2	0.009
hs-CRP, mg/dL	7.0 ± 21.0	9.1 ± 18.5	7.5 ± 10.9	0.87
NT-proBNP, pg/mL	948 (515-1,467)	629 (337-1,430)	614 (393-1,610)	0.50
Log NT-proBNP	6.8 ± 0.9	6.5 ± 1.1	6.6 ± 1.0	0.65
LVEF, %	57.9 ± 6.1	59.0 ± 6.9	60.3 ± 4.9	0.24
LAV, mL	92.7 ± 31.2	86.7 ± 32.7	84.0 ± 33.0	0.51
LAD, mm	46.9 ± 6.1	45.9 ± 5.0	44.9 ± 6.1	0.35
LVM, g/m ²	194.1 ± 59.5	193.7 ± 52.5	190.5 ± 52.0	0.96
Lateral E/e'	12.6 ± 6.2	11.6 ± 4.2	9.7 ± 3.5	0.037
PASP, mm Hg	38.2 ± 12.2	36.5 ± 11.3	37.8 ± 16.0	0.88
PWV, m/s	11.9 ± 1.7	10.9 ± 1.5	11.2 ± 1.4	0.019
UACR, mg/g	22.9 (10.4-82.8)	23.6 (8.2-107.1)	30.3 (11.3-89.7)	0.79
Log UACR	3.4 ± 1.6	3.6 ± 1.8	3.6 ± 1.6	0.86
USCR, mmol/g	120 (57-224)	147 (64-248)	86 (35-200)	0.39

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TABLE 1 Continued

	Renin T1	Renin T2	Renin T3	P Value
Urine sodium, mmol/L	82.1 ± 32.3	85.2 ± 41.2	62.2 ± 29.9	0.022
Urine potassium, mmol/L	44.6 ± 27.0	42.3 ± 24.0	48.1 ± 30.1	0.66
Log USCR	4.6 ± 0.9	4.8 ± 1.2	4.5 ± 1.2	0.74
Log renin	2.3 ± 0.8	3.4 ± 0.2	4.8 ± 0.9	<0.001
Aldosterone, pg/mL	51.2 (41.4-94.6)	66.5 (38.7-121.0)	94.4 (44.3-117.0)	0.19
Log aldosterone	4.1 ± 0.5	4.2 ± 0.6	4.4 ± 0.7	0.17
ARR	5.0 (3.0-10.6)	2.3 (1.3-4.1)	0.7 (0.3-1.3)	<0.001
Log ARR	1.8 ± 1.0	0.8 ± 0.7	-0.5 ± 1.0	<0.001

Values are median (25th-75th percentile), n (%), or mean ± SD, unless otherwise indicated.

ACEI = angiotensin converting enzyme inhibitor; AF = atrial fibrillation; ARB = angiotensin receptor blocker; ARR = aldosterone-to-renin ratio; BMI = body mass index; BP = blood pressure; CCB = calcium-channel blocker; CKD = chronic kidney disease; DBP = diastolic blood pressure; ECG = electrocardiogram; eGFR = estimated glomerular filtration rate; HbA_{1c} = glycosylated hemoglobin; HF = heart failure; HFH = heart failure hospitalization; hs-CRP = high-sensitivity C-reactive protein; JVD = jugular venous distention; GLP1-RA = glucagon-like peptide-1 receptor agonist; LAD = left atrial diameter; LAV = left atrial volume; LVEF = left ventricular ejection fraction; LVM = left ventricular mass; MI = myocardial infarction; NT-proBNP = N-terminal pro-B-type natriuretic peptide; PASP = pulmonary artery systolic pressure; PWV = pulse wave velocity; SBP = systolic blood pressure; T = tertile; UACR = spot urinary albumin-to-creatinine ratio; USCR = spot urinary sodium-to-creatinine ratio.

were: T1 (n = 36), 32.8 pg/mL (25th-75th percentile: 27.7-42.2 pg/mL); T2 (n = 35), 69.7 pg/mL (25th-75th percentile: 55.1-90.6 pg/mL); and T3 (n = 35), 128 pg/mL (25th-75th percentile: 115-158 pg/mL). Compared with patients with lower aldosterone levels (T1 and T2), those with higher levels (T3) had been more frequently hospitalized for HF in the previous year, had lower eGFR, lower serum sodium levels, lower spot urine sodium excretion, and higher spot urine potassium excretion. No statistically significant differences in age, sex, background HF medication, diabetes status, serum potassium, or serum renin levels were observed (Supplemental Table 1).

The median ARR at baseline was 2.3 (25th-75th percentile: 0.9 to 4.4). Because plasma aldosterone levels did not differ significantly across renin tertiles, ARR tertiles exhibited a similar pattern to that observed for plasma renin (Supplemental Table 2). The correlation between plasma renin and aldosterone was poor and not statistically significant ($\rho = 0.15$; $P = 0.12$) (Supplemental Figure 1).

INFLUENCE OF TREATMENT SEQUENCES ON PLASMA RENIN AND ALDOSTERONE LEVELS. Compared with baseline, dapagliflozin alone induced a modest increase in plasma renin levels (geometric mean ratio [gMratio] renin change = 1.32; 95% CI: 1.07-1.62; $P = 0.036$). Renin increases were more pronounced in treatment sequences including spironolactone, particularly with the spironolactone/dapagliflozin combination (gMratio renin change = 3.53; 95% CI: 2.62-4.74; $P < 0.001$), followed by spironolactone alone (gMratio renin change = 2.58; 95% CI: 1.71-3.91; $P < 0.001$). Consequently, relative to dapagliflozin alone, sequences including spironolactone led to significantly higher plasma renin levels, particularly with the spironolactone/dapagliflozin combination

resulting in a >2.5-fold increase (gMratio renin difference = 2.65; 95% CI: 2.06-3.41; $P < 0.001$) (Figure 1A, Table 2).

Compared with baseline, dapagliflozin alone led to a modest increase in plasma aldosterone levels (gMratio aldosterone change = 1.18; 95% CI: 1.07-1.29; $P = 0.016$). In line with renin findings, aldosterone increases were more pronounced in sequences including spironolactone, particularly with the spironolactone/dapagliflozin combination (gMratio aldosterone change = 2.10; 95% CI: 1.83-2.40; $P < 0.001$), followed by spironolactone alone (gMratio aldosterone change = 1.89; 95% CI: 1.51-2.36; $P < 0.001$). Accordingly, compared with dapagliflozin alone, the spironolactone/dapagliflozin combination resulted in >1.5-fold higher plasma aldosterone levels (gMratio aldosterone difference = 1.77; 95% CI: 1.56-2.02; $P < 0.001$) (Figure 1B, Table 2).

Because plasma renin increased more than aldosterone with spironolactone, the ARR was lower with the spironolactone/dapagliflozin combination than with dapagliflozin alone (Table 2).

TREATMENT EFFECT MODIFICATION BY PLASMA RENIN AND ALDOSTERONE LEVELS. Plasma renin levels, but not aldosterone, modified the BP effects of treatment sequences including spironolactone, particularly the spironolactone/dapagliflozin combination, compared with dapagliflozin alone. Effect modification was most pronounced for SBP and DBP (interaction P trend < 0.05 for both). Among patients with low baseline plasma renin (T1), the spironolactone/dapagliflozin combination reduced SBP by -10.9 mm Hg (95% CI: -16.4 to -5.4 mm Hg) vs dapagliflozin alone. There was no significant reduction in SBP observed at higher baseline plasma renin levels: T2, -5.6 mm Hg (95% CI: -11.3 to 0.02 mm Hg);

and T3, 1.5 mm Hg (95% CI: −4.1 to 7.0 mm Hg); interaction *P* trend = 0.007. There was a similar pattern for DBP (Figure 2, Table 3).

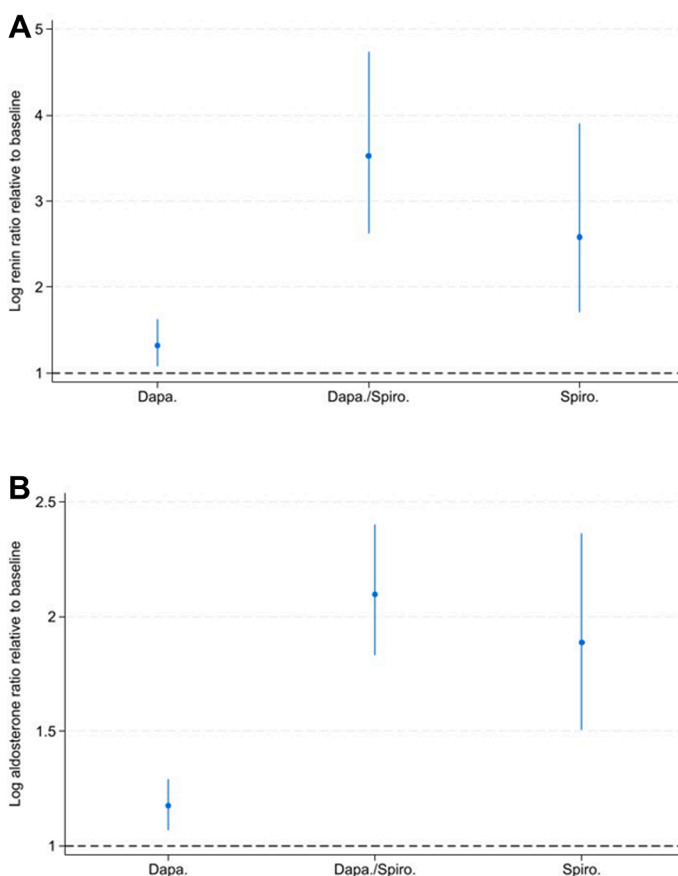
Lateral E/e' showed a greater reduction from baseline with both dapagliflozin and spironolactone/dapagliflozin in patients with low renin levels (T1), with no significant changes at higher renin tertiles (T2 and T3; interaction *P* trend < 0.05). Similarly, PWV tended to decrease more at low renin levels (T1) with both dapagliflozin and spironolactone/dapagliflozin, with no significant changes at higher renin levels (T2 and T3; interaction *P* trend = 0.063).

Plasma renin levels did not influence the effect of the treatment sequences on NT-proBNP (interaction *P* trend = 0.25). Similarly, renin did not significantly modify eGFR decline (interaction *P* trend = 0.72). Nevertheless, eGFR decline tended to be more pronounced at low renin levels (T1) across all treatment sequences, particularly those including spironolactone. Serum potassium increased more with treatment sequences including spironolactone, without significant modulation by plasma renin (interaction *P* trend = 0.74) (Table 3). Left atrial diameter reduction was greater at lower renin levels (T1 and T2) with both dapagliflozin and spironolactone/dapagliflozin sequences (interaction *P* trend = 0.027). Although the interaction was not statistically significant (interaction *P* trend = 0.79), left ventricular mass also appeared to decrease more with spironolactone/dapagliflozin and spironolactone alone at low renin levels (T1). Spot urine sodium excretion decreased from baseline only at low renin levels (T1) with spironolactone/dapagliflozin, again with no statistically significant treatment-by-renin interaction (interaction *P* trend = 0.22) (Supplemental Table 3).

Despite the absence of statistically significant interactions, both SBP and DBP reductions with the spironolactone/dapagliflozin tended to be greater at lower aldosterone levels (T1 and T2) than at higher levels (T3) (interaction *P* trend = 0.062 for SBP and interaction *P* trend = 0.085 for DBP). eGFR decline and urinary albumin-to-creatinine ratio reduction with spironolactone/dapagliflozin appeared more pronounced at lower aldosterone levels (T1 and T2), compared with higher levels (T3) (interaction *P* trend = 0.10 for both). In contrast, body weight reduction with spironolactone/dapagliflozin tended to be greater at higher aldosterone levels (T3) compared with lower aldosterone levels (T1 and T2) (interaction *P* trend = 0.086) (Supplemental Table 3).

Despite lower baseline values, spot urine sodium excretion increased progressively across aldosterone tertiles (interaction *P* trend = 0.019) (Supplemental

FIGURE 1 Effect of Treatment Sequences on Plasma Renin and Aldosterone Levels



Plasma renin (A) and aldosterone (B) changes according to treatment sequences are expressed as the geometric mean ratio with 95% CIs. Dapa. = dapagliflozin; Spiro. = spironolactone.

Table 3). Plasma aldosterone did not influence the effects of treatment sequences on NT-proBNP, echocardiographic parameters, PWV, or serum potassium (all interaction *P* trend > 0.10). The ARR exhibited a pattern similar to that of plasma renin, consistent with renin being the main driver of effect modification on BP, echocardiographic measures, and PWV.

VARIABLES ASSOCIATED WITH LOW RENIN LEVELS. In an exploratory analysis, the baseline variables associated with low renin levels (T1; ≤ 21 mU/L) were BP >130/80 mm Hg, higher serum sodium, greater lateral E/e', and lower uric acid (model area under the curve = 0.68) (Supplemental Table 4).

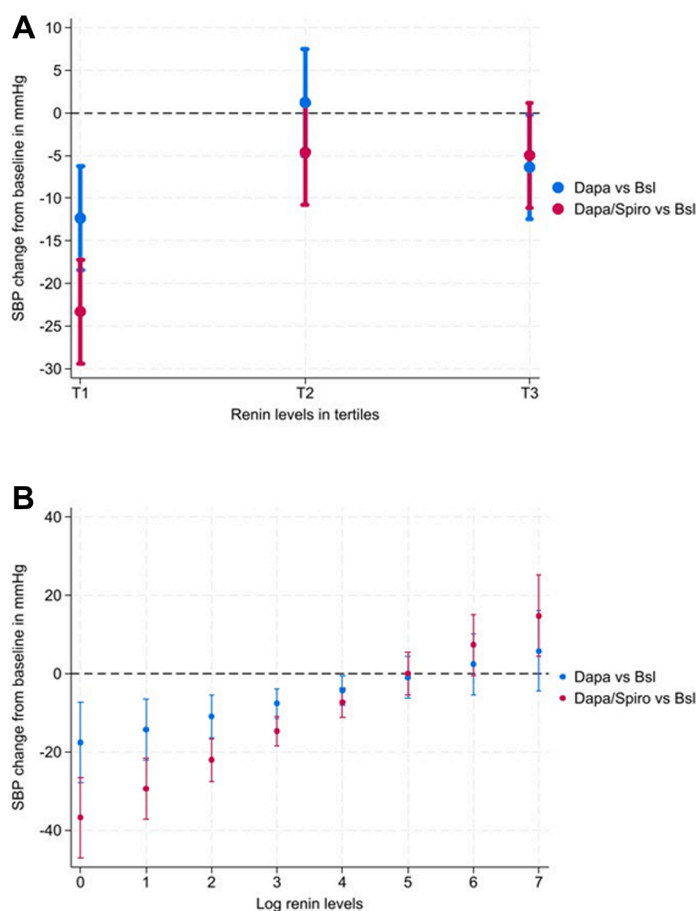
SUBGROUP ANALYSES. The treatment-by-renin interaction for SBP exhibited a similar pattern regardless of baseline hypertension status or the number of

TABLE 2 Effect of Treatment Sequences on the Plasma Renin, Aldosterone, and ARR Levels

	BL (n = 106)	Geometric Mean Ratio (95% CI)		
		DAPA (n = 103)	SPL/DAPA (n = 103)	SPL (n = 41)
Renin, mU/L	3.5 ± 1.3; ^a 30 (18-62) ^b	vs BL ^c : 1.32 (1.07-1.62) vs DAPA: Ref.	vs BL ^c : 3.53 (2.62-4.74) vs DAPA ^c : 2.65 (2.06-3.41)	vs BL ^c : 2.58 (1.71-3.91) vs DAPA ^c : 2.14 (1.41-3.24)
Aldosterone, pg/mL	4.2 ± 0.6; ^a 68 (42-115) ^b	vs BL ^c : 1.18 (1.07-1.29) vs DAPA: Ref.	vs BL ^c : 2.10 (1.83-2.40) vs DAPA ^c : 1.77 (1.56-2.02)	vs BL ^c : 1.89 (1.51-2.36) vs DAPA ^c : 1.65 (1.29-2.09)
ARR	0.7 ± 1.3; ^a 2.3 (0.9-4.4) ^b	vs BL: 0.89 (0.73-1.08) vs DAPA: Ref.	vs BL ^c : 0.60 (0.45-0.79) vs DAPA ^c : 0.67 (0.52-0.86)	vs BL: 0.74 (0.47-1.17) vs DAPA: 0.79 (0.52-1.21)

^aMean ± SD of log-transformed data. ^bMedian (25th-75th percentile). ^cP < 0.05.

ARR = aldosterone-to-renin ratio; BL = baseline; DAPA = dapagliflozin; Ref. = Reference; SPL = spironolactone.

FIGURE 2 Influence of Plasma Renin Levels at Baseline on the Effect of Treatment Sequences on SBP

The y-axis represents the change in systolic blood pressure (SBP) along with the respective 95% CIs. The x-axis represents renin levels in tertiles (A) and in log-transformed continuous values (B). Tertiles of plasma renin in mU/L median: tertile 1 (T1) = 12.4 (25th-75th percentile: 6.7-18.0); tertile 2 (T2) = 30.0 (25th-75th percentile: 24.0-36.4); and tertile 3 (T3) = 96.1 (25th-75th percentile: 61.6-211.0). The treatment response with the Dapa./Spiro. combination vs Dapa. alone becomes more pronounced at renin levels ≤3 log units (ie, <21 mU/L = T1 values). Abbreviations as in Figure 1.

antihypertensive medications (interaction *P* trend > 0.10 for both) (Supplemental Table 5).

A summary of the findings is displayed in the Central Illustration.

DISCUSSION

In this analysis of patients with HFpEF enrolled in the SOGALDI-PEF trial, we examined how baseline renin and aldosterone levels influence the effects of dapagliflozin and the dapagliflozin/spironolactone combination on BP, echocardiographic parameters, and blood and urine biomarkers, and identified several findings with potential clinical and research implications. First, baseline renin and aldosterone levels detected 2 distinct HFpEF phenotypes. Patients with low baseline renin levels (ie, ≤21 mU/L) had higher BP, more frequent peripheral edema, and higher lateral E/e', PWV, and serum sodium, despite a tendency toward higher eGFR, lower urea, and lower uric acid. In contrast, higher baseline aldosterone (ie, ≥98 pg/mL) was associated with lower eGFR, lower serum sodium and spot urine sodium excretion, and higher spot urine potassium excretion, with no major differences in BP, echocardiographic parameters, PWV, or peripheral edema observed across aldosterone levels. Importantly, baseline renin and aldosterone were independent (ie, uncorrelated), supporting the presence of distinct renin-low and aldosterone-high phenotypes.

Second, spironolactone-containing sequences produced a marked neurohumoral feedback response. Compared with dapagliflozin, which led to only modest increases in renin and aldosterone,¹⁹ the spironolactone/dapagliflozin combination increased plasma renin levels by >2.5-fold and aldosterone >1.5-fold, consistent with activation of the renin-angiotensin-aldosterone system after MR blockade.²⁰ Third, low renin identified a subgroup with a substantially greater hemodynamic and

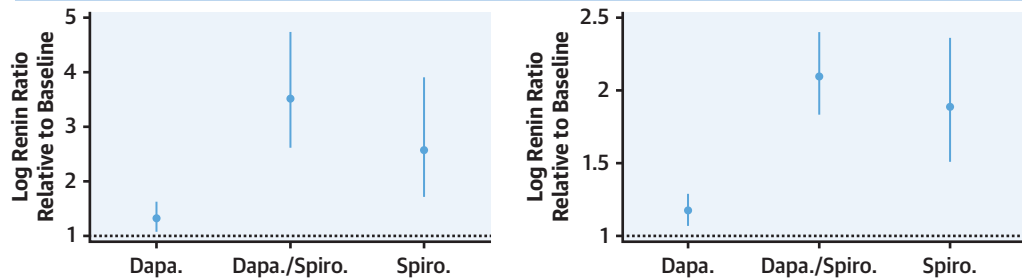
TABLE 3 Influence of Plasma Renin Levels on the Effect of Treatment Sequences

	BL (n = 106)	DAPA (n = 103)	SPL/DAPA (n = 103)	SPL (n = 41)	Interaction P Trend Value ^a
SBP changes by treatment sequences, mm Hg					
Renin T1 (n = 36); 12.4 mU/L (95% CI: 6.7-18.0 mU/L)	139.7 ± 17.2	vs BL: −12.3 (−18.5 to −6.2) vs DAPA: Ref.	vs BL: −23.3 (−29.4 to −17.2) vs DAPA: −10.9 (−16.4 to −5.4)	vs BL: −19.9 (−26.6 to −10.1) vs DAPA: −7.6 (−16.5 to 1.3)	0.007
Renin T2 (n = 35); 30.0 mU/L (95% CI: 24.0-36.4 mU/L)	132.2 ± 18.8	vs BL: 1.2 (−5.0 to 7.5) vs DAPA: Ref.	vs BL: −4.6 (−10.8 to 1.6) vs DAPA: −5.6 (−11.3 to 0.02)	vs BL: −2.5 (−11.1 to 5.9) vs DAPA: −3.2 (−11.1 to 4.7)	
Renin T3 (n = 35); 96.1 mU/L (95% CI: 61.6-211 mU/L)	128.1 ± 19.6	vs BL: −6.4 (−12.5 to −0.2) vs DAPA: Ref.	vs BL: −5.0 (−11.2 to 1.2) vs DAPA: 1.5 (−4.1 to 7.0)	vs BL: −3.5 (−11.4 to 4.4) vs DAPA: 2.5 (−4.7 to 9.6)	
DBP changes by treatment sequences, mm Hg					
Renin T1 (n = 36); 12.4 mU/L (95% CI: 6.7-18.0 mU/L)	76.3 ± 8.8	vs BL: −4.7 (−8.4 to −1.0) vs DAPA: Ref.	vs BL: −10.9 (−14.6 to −7.2) vs DAPA: −6.3 (−10.2 to −2.4)	vs BL: −13.7 (−19.6 to −7.8) vs DAPA: −8.9 (−15.1 to −2.7)	0.006
Renin T2 (n = 35); 30.0 mU/L (95% CI: 24.0-36.4 mU/L)	73.8 ± 11.6	vs BL: −1.2 (−5.0 to −2.6) vs DAPA: Ref.	vs BL: −1.2 (−4.9 to 2.6) vs DAPA: 0.1 (−3.8 to 4.2)	vs BL: −1.7 (−6.8 to 3.5) vs DAPA: −0.2 (−5.7 to 5.3)	
Renin T3 (n = 35); 96.1 mU/L (95% CI: 61.6-211 mU/L)	69.3 ± 10.2	vs BL: −1.1 (−4.8 to 2.6) vs DAPA: Ref.	vs BL: −0.6 (−4.3 to 3.2) vs DAPA: 0.4 (−3.5 to 4.4)	vs BL: 2.6 (−2.2 to 7.4) vs DAPA: 3.8 (−1.1 to 8.8)	
Lateral E/e' changes by treatment sequences					
Renin T1 (n = 36); 12.4 mU/L (95% CI: 6.7-18.0 mU/L)	12.6 ± 6.2	vs BL: −1.6 (−2.9 to −0.3) vs DAPA: Ref.	vs BL: −1.6 (−2.8 to −0.3) vs DAPA: 0.0 (−1.3 to 1.3)	vs BL: 2.6 (0.5 to 4.7) vs DAPA: 4.4 (2.2 to 6.6)	0.013
Renin T2 (n = 35); 30.0 mU/L (95% CI: 24.0-36.4 mU/L)	11.6 ± 4.2	vs BL: −0.3 (−1.6 to 1.0) vs DAPA: Ref.	vs BL: 0.3 (−1.0 to 1.6) vs DAPA: 0.7 (−0.7 to 2.0)	vs BL: −0.1 (−2.0 to 1.7) vs DAPA: 0.5 (−1.4 to 2.4)	
Renin T3 (n = 35); 96.1 mU/L (95% CI: 61.6-211 mU/L)	9.7 ± 3.5	vs BL: −0.3 (−1.6 to 1.0) vs DAPA: Ref.	vs BL: −0.1 (−1.4 to 1.2) vs DAPA: 0.2 (−1.2 to 1.5)	vs BL: −0.5 (−2.2 to 1.2) vs DAPA: −0.2 (−2.0 to 1.5)	
PWV changes by treatment sequences, m/s					
Renin T1 (n = 36); 12.4 mU/L (95% CI: 6.7-18.0 mU/L)	11.9 ± 1.7	vs BL: −0.5 (−0.7 to −0.2) vs DAPA: Ref.	vs BL: −0.8 (−1.0 to −0.5) vs DAPA: −0.3 (−0.6 to −0.1)	vs BL: −0.6 (−1.0 to −0.2) vs DAPA: −0.2 (−0.6 to 0.2)	0.063
Renin T2 (n = 35); 30.0 mU/L (95% CI: 24.0-36.4 mU/L)	10.9 ± 1.5	vs BL: 0.1 (−0.1 to 0.4) vs DAPA: Ref.	vs BL: −0.1 (−0.3 to 0.2) vs DAPA: −0.2 (−0.5 to 0.0)	vs BL: 0.1 (−0.3 to 0.4) vs DAPA: −0.0 (−0.4 to 0.3)	
Renin T3 (n = 35); 96.1 mU/L (95% CI: 61.6-211 mU/L)	11.2 ± 1.4	vs BL: −0.2 (−0.5 to 0.0) vs DAPA: Ref.	vs BL: −0.1 (−0.4 to 0.1) vs DAPA: 0.1 (−0.2 to 0.3)	vs BL: −0.2 (−0.5 to 0.2) vs DAPA: 0.0 (−0.3 to 0.3)	
NT-proBNP changes by treatment sequences in log units					
Renin T1 (n = 36); 12.4 mU/L (95% CI: 6.7-18.0 mU/L)	6.8 ± 0.9	vs BL: −0.2 (−0.4 to 0.1) vs DAPA: Ref.	vs BL: −0.3 (−0.5 to −0.1) vs DAPA: −0.1 (−0.3 to 0.1)	vs BL: −0.1 (−0.4 to 0.2) vs DAPA: 0.1 (−0.3 to 0.3)	0.25
Renin T2 (n = 35); 30.0 mU/L (95% CI: 24.0-36.4 mU/L)	6.5 ± 1.1	vs BL: 0.0 (−0.2 to 0.2) vs DAPA: Ref.	vs BL: −0.2 (−0.4 to 0.0) vs DAPA: −0.1 (−0.3 to 0.1)	vs BL: −0.1 (−0.4 to 0.2) vs DAPA: −0.1 (−0.4 to 0.2)	
Renin T3 (n = 35); 96.1 mU/L (95% CI: 61.6-211 mU/L)	6.6 ± 1.0	vs BL: 0.0 (−0.2 to 0.2) vs DAPA: Ref.	vs BL: −0.1 (−0.3 to 0.1) vs DAPA: −0.1 (−0.3 to 0.1)	vs BL: −0.2 (−0.5 to 0.0) vs DAPA: −0.2 (−0.5 to 0.0)	
eGFR changes by treatment sequences, mL/min/1.73 m²					
Renin T1 (n = 36); 12.4 mU/L (95% CI: 6.7-18.0 mU/L)	74.1 ± 20.3	vs BL: −6.9 (−10.7 to −3.0) vs DAPA: Ref.	vs BL: −14.1 (−17.9 to −10.3) vs DAPA: −7.3 (−10.7 to −3.9)	vs BL: −11.5 (−17.7 to −5.3) vs DAPA: −4.1 (−9.8 to 1.6)	0.72
Renin T2 (n = 35); 30.0 mU/L (95% CI: 24.0-36.4 mU/L)	70.2 ± 22.4	vs BL: −2.0 (−6.0 to 1.9) vs DAPA: Ref.	vs BL: −7.5 (−11.3 to −3.6) vs DAPA: −5.2 (−8.8 to −1.7)	vs BL: −3.7 (−9.1 to 1.7) vs DAPA: −1.5 (−6.5 to 3.5)	
Renin T3 (n = 35); 96.1 mU/L (95% CI: 61.6-211 mU/L)	63.9 ± 22.6	vs BL: −1.5 (−5.3 to 2.4) vs DAPA: Ref.	vs BL: −7.8 (−11.7 to −3.9) vs DAPA: −6.4 (−9.9 to −3.0)	vs BL: −3.6 (−8.6 to −1.4) vs DAPA: −2.3 (−6.8 to 2.2)	
Serum potassium changes by treatment sequences, mmol/L					
Renin T1 (n = 36); 12.4 mU/L (95% CI: 6.7-18.0 mU/L)	4.3 ± 0.4	vs BL: 0.0 (−0.2 to 0.2) vs DAPA: Ref.	vs BL: 0.4 (0.2 to 0.5) vs DAPA: 0.4 (0.2 to 0.5)	vs BL: 0.3 (0.0 to 0.6) vs DAPA: 0.3 (0.1 to 0.6)	0.74
Renin T2 (n = 35); 30.0 mU/L (95% CI: 24.0-36.4 mU/L)	4.2 ± 0.5	vs BL: 0.2 (0.0 to 0.4) vs DAPA: Ref.	vs BL: 0.5 (0.3 to 0.6) vs DAPA: 0.3 (0.1 to 0.4)	vs BL: 0.1 (−0.2 to 0.3) vs DAPA: −0.1 (−0.4 to 0.1)	
Renin T3 (n = 35); 96.1 mU/L (95% CI: 61.6-211 mU/L)	4.4 ± 0.5	vs BL: −0.1 (−0.3 to 0.1) vs DAPA: Ref.	vs BL: 0.2 (0.1 to 0.4) vs DAPA: 0.3 (0.2 to 0.5)	vs BL: 0.1 (−0.1 to 0.3) vs DAPA: 0.2 (0.0 to 0.4)	

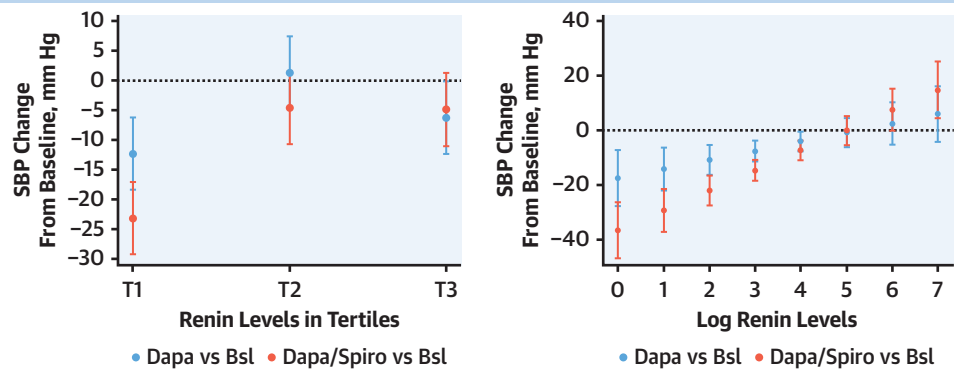
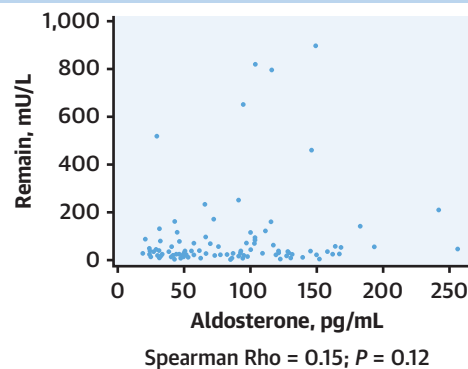
^aStatistically significant interactions present both using renin as a continuous and categorical (tertile; ie interaction P trend test) covariate.Ref. = Reference; other abbreviations as in [Tables 1 and 2](#).

cardiovascular response to treatment. The BP-lowering effect of spironolactone-containing sequences was more pronounced at low renin levels (≤ 21 mU/L), with the spironolactone/dapagliflozin

combination in this subgroup almost doubling the BP reduction observed with dapagliflozin alone, while it did not significantly reduce BP in patients with higher baseline renin (>21 mU/L). Reductions in

CENTRAL ILLUSTRATION Summary of Renin and Aldosterone Findings in SOGALD-PEF**SOGALDI-PEF (N = 108)**
Cross-Over Trial With 3 Sequences: Dapagliflozin, Spironolactone, Dapagliflozin/Spironolactone**Renin and Aldosterone Measured at Baseline and at Each Sequence****Spironolactone Increased Renin and Aldosterone Levels: Feedback Mechanism**

Aldosterone levels did not seem to influence the response to dapagliflozin/spironolactone.

Blood Pressure Reduction Was More Pronounced With Dapagliflozin/Spironolactone vs Dapagliflozin at Low Renin Levels (≤ 21 mU/L)**Renin and Aldosterone Levels were Poorly Correlated**

Ferreira JP, et al. JACC Heart Fail. 2026;■(■):103111.

In SOGALDI-PEF, a 3-sequence crossover trial with dapagliflozin (Dapa.), spironolactone (Spiro.), and the Dapa./Spiro. combination, the sequences including Spiro. led to an increase in aldosterone and renin levels due a known feedback mechanism consequently to mineralocorticoid receptor blockade. At baseline (Bsl.), before any study drug administration, renin and aldosterone were poorly correlated. Contrary to aldosterone, which was not associated with treatment response, low renin levels predicted a more pronounced blood pressure reduction with the sequences containing Spiro.

lateral E/e' and PWV with dapagliflozin and spironolactone/dapagliflozin were also more pronounced at low renin levels, whereas renin did not significantly influence changes in NT-proBNP, eGFR, or serum potassium.

Fourth, aldosterone had limited incremental value beyond renin for predicting treatment response. Although patients with higher baseline aldosterone levels (≥ 98 pg/mL) had a more adverse renal and sodium-handling profile, baseline aldosterone levels did not significantly modify the effects of any treatment sequence on BP, NT-proBNP, echocardiographic parameters, PWV, or serum potassium. Neither aldosterone nor the ARR improved discrimination of hemodynamic or cardiovascular responses beyond renin alone, suggesting that in HFpEF, aldosterone behaves primarily as a marker of altered sodium handling and renal dysfunction rather than a determinant of treatment benefit. Supporting our findings, in patients with resistant hypertension, spironolactone reduced BP and improved endothelial function and echocardiographic parameters related to diastolic dysfunction and left ventricular hypertrophy independently of aldosterone excess or escape.²¹ Furthermore, aging alters the intracellular redox state, increasing the generation of reactive oxygen species and oxidized glutathione and inhibiting 11β hydroxysteroid dehydrogenase type 2; these conditions increase MR expression and facilitate the action of cortisol to act as an MR agonist. Therefore, in elderly populations, such as the population included in SOGALDI-PEF, MR expression may be increased and activated by cortisol even under normal or minimally elevated aldosterone levels. In such conditions, low renin may be a more reliable biomarker of MRA effectiveness.²²

These findings are concordant with, and extend, previous work in hypertension and HF risk states. In PATHWAY2 (Optimum Treatment for Drug-Resistant Hypertension; [NCT02369081](#)), which compared four 12-week treatment sequences (spironolactone, bisoprolol, doxazosin, and placebo) in a crossover design among patients with treatment-resistant hypertension (mean SBP: 148 mm Hg; age: 61 years; eGFR: 91 mL/min/1.73 m²; serum potassium: 4.1 mmol/L), spironolactone was the most effective agent for lowering BP.⁵ However, the SBP reduction with spironolactone was much more pronounced at the lower end of baseline renin distribution, exceeding 15 mm Hg vs baseline and 10 mm Hg vs the other agents. Median baseline renin was 34 mU/L in PATHWAY2, similar to that in SOGALDI-PEF (approximately 30 mU/L), and in both studies renin was measured as direct renin concentrations in

morning blood samples without withdrawal of background therapy. In HOMAGE (Bioprofiling Response to Mineralocorticoid Receptor Antagonists for the Prevention of Heart Failure; [NCT02556450](#)), which enrolled patients at risk of developing HF (mean SBP: 140 mm Hg; age: 73 years; eGFR: 73 mL/min/1.73 m²; serum potassium: 4.3 mmol/L), spironolactone given for 9 months on top of standard care reduced SBP by about 10 mm Hg overall; reductions exceeded 15 mm Hg at the lower end of baseline renin. In HOMAGE, renin was quantified by using an Olink platform, and these relative concentrations cannot be directly converted to direct renin concentrations.⁶

The larger BP-lowering effect of spironolactone at the lower end of the baseline renin distribution suggests that sodium retention is an important pathophysiological driver across hypertension, HF risk states, and established HFpEF.²³ Other studies have likewise indicated that patients with lower baseline renin levels tend to exhibit larger SBP reductions in response to MRAs and aldosterone synthase inhibitors. However, differences in study populations, drugs and doses, renin assays, and renin cutoffs limit a robust, trial-level replication of renin-dependent modification of BP response.²⁴

It is important to note that patients with low baseline renin levels (≤ 21 mU/L) had higher SBP, averaging 140 mm Hg, whereas those with higher renin levels had SBP of approximately 130 mm Hg or lower. However, the greater SBP reduction observed in the low-renin group cannot be attributed simply to regression to the mean. In these patients, dapagliflozin alone already produced a marked BP reduction, consistent with its short-term diuretic effect,^{19,25} yet the spironolactone/dapagliflozin combination almost doubled this BP lowering in the same low-renin subgroup. In contrast, both dapagliflozin and the spironolactone/dapagliflozin combination had minimal BP effects in patients with higher baseline renin. Another relevant observation is that dapagliflozin and the spironolactone/dapagliflozin combination lowered BP without increasing spot urine sodium excretion, suggesting an increase in urine volume rather than natriuresis, in line with the absence of differences in urinary sodium excretion after adjustment for urinary creatinine and with previous mechanistic data.²² Furthermore, the spironolactone doses used in SOGALDI-PEF (< 20 mg/d on average) are not natriuretic doses and therefore are not expected to substantially enhance natriuresis.²⁶

Beyond BP, patients with low baseline renin exhibited greater reductions in lateral E/e', PWV, and left atrial diameter with both the dapagliflozin and

spironolactone/dapagliflozin combination, supporting additional cardiac and vascular benefits in this subgroup. Conversely, patients with HF and high renin levels may have an attenuated response to renin-angiotensin-aldosterone system inhibition and a worse prognosis, a phenomenon that was not observed with high aldosterone levels.²⁰

Baseline renin levels did not appear to substantially influence the magnitude of eGFR decline or potassium increase with the spironolactone/dapagliflozin combination. In fact, eGFR decline was most marked in patients with low renin levels, probably because they had higher baseline eGFR and experienced greater BP reduction, as short-term eGFR changes are closely linked to hemodynamic effects.²⁷ NT-proBNP level reduction with spironolactone/dapagliflozin was also not influenced by renin levels. This may be explained by the relatively small magnitude NT-proBNP reduction in the overall population (approximately 11%) with a marked impact of atrial fibrillation on NT-proBNP levels as 56% of the SOGALDI-PEF population had atrial fibrillation at baseline; these patients did not experience a reduction in NT-proBNP levels with treatment.¹

Beyond increased sodium retention and extracellular content, a significant proportion of patients with low-renin phenotype are thought to have primary aldosteronism, often in subclinical forms with normal or only mildly elevated aldosterone levels. Nonetheless, in SOGALDI-PEF, renin and aldosterone were not correlated, suggesting that classical renin-aldosterone interdependence was not predominant in this setting.²⁸ BP, echocardiographic parameters, and PWV did not vary significantly across baseline aldosterone strata, whereas eGFR, serum sodium, and spot urinary sodium excretion were lower at higher aldosterone levels. Serum potassium levels also tended to be higher with increasing aldosterone, although differences were not statistically significant. This profile is consistent with the established actions of aldosterone to increase distal sodium reabsorption and potassium excretion.²⁹ Accordingly, in patients with high baseline aldosterone levels, both dapagliflozin and the spironolactone/dapagliflozin combination increased spot urine sodium excretion while reducing urinary potassium excretion. Because these changes were also observed with dapagliflozin alone, they do not seem to depend solely on aldosterone blockade and instead suggest that high aldosterone marks a sodium-avid state that can be partially reversed by dapagliflozin, spironolactone, and, particularly, their combination. Although high aldosterone likely identifies a high-risk subgroup who benefits from MRA therapy after

a myocardial infarction complicated with systolic dysfunction,³⁰ this pattern may not be observed in patients with chronic HFpEF.²

STUDY LIMITATIONS. First, renin and aldosterone were measured in morning blood samples without stopping angiotensin-converting enzyme inhibitors/angiotensin receptor blockers, diuretics, or beta-blockers. Although these drugs can alter renin and aldosterone levels, our findings therefore reflect patients with HFpEF on contemporary background therapy and are directly applicable to routine clinical practice. Second, the trial was not powered for formal interaction testing. Nevertheless, the consistent treatment-by-renin interactions for SBP and DBP provide internal coherence, and concordant renin findings in PATHWAY-2 and HOMAGE provide external coherence. Third, the absence of correlation between renin and aldosterone may appear counterintuitive. However, in this HFpEF population, renin levels were lower in the subgroup with a phenotype characterized by sodium overload and high BP with preserved renal function, whereas aldosterone levels were higher in patients with impaired renal function, low serum sodium, and higher serum potassium. These findings suggest that these hormones reflect partly independent pathophysiological axes. Moreover, the rather small sample size and lack of “hard” outcomes preclude from establishing associations between renin, aldosterone, and outcomes. The weak associations between aldosterone and MRA treatment response may be partially attributed to the limited statistical power of this study. Finally, because this study was not designed or powered to evaluate hospitalizations or mortality, these observations should be further validated in larger outcome trials.

CONCLUSIONS

In patients with HFpEF, low plasma renin levels (≤ 21 mU/L) identify a sodium overload and high BP phenotype and predict a substantially greater BP reduction with the spironolactone/dapagliflozin combination than with dapagliflozin. In this low-renin subgroup, the spironolactone/dapagliflozin combination also seemed to exert more favorable effects on cardiovascular structure and function. In comparison, baseline aldosterone levels did not predict treatment response. Whether renin can be used to stratify treatment benefit on hard outcomes should be addressed in future outcome trials.

DATA AVAILABILITY Anonymized data will be shared approximately 1 year after the main publication on the Vivli platform.

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PERSPECTIVES

COMPETENCY IN MEDICAL KNOWLEDGE: This analysis from SOGALDI-PEF suggests that low renin levels might be used to predict the response to spironolactone with and without dapagliflozin combination in terms of BP and PWV reduction as well as mitigation of adverse cardiac remodeling.

TRANSLATIONAL OUTLOOK: This study supports continuing research on the renin-angiotensin-aldosterone system in HFpEF and hypertension, as it challenges the current view that aldosterone elevation is the main driver of sodium retention and adverse outcomes.

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KEY WORDS aldosterone, dapagliflozin, heart failure with preserved ejection fraction, renin, spironolactone

APPENDIX For a list of the SOGALDI-PEF investigators as well as a supplemental figure and tables, please see the online version of this paper.